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Image forming apparatus that announces abnormality of sheet

Abstract

An image forming apparatus includes an image forming device, a sheet transport mechanism, a sheet abnormality detection device, an output tray, an escape tray, a notification device, and a control device. The escape tray is provided separately from the output tray, and receives the sheet, from which abnormality has been detected by the sheet abnormality detection device, and delivered by the sheet transport mechanism. The control device causes the image forming device to again form, on another sheet, a same image formed on a sheet, from which the sheet abnormality has been detected, and delivered to the escape tray, and to continue with the image forming operation on subsequent sheets, and decides, according to a predetermined condition, whether to allow the notification device to announce the sheet abnormality.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application claims priority to Japanese Patent Application No. 2022-203233 filed on Dec. 20, 2022, the entire contents of which are incorporated by reference herein.

BACKGROUND

(2) The present disclosure relates to an image forming apparatus, and in particular to an image forming apparatus including an image forming device that forms an image on a sheet delivered thereto.

(3) Among existing sheet transport devices installed in image forming apparatuses, some are configured to notify a user of occurrence of abnormality of a sheet, thereby allowing the user to be immediately made aware of the occurrence of abnormality.

SUMMARY

(4) The disclosure proposes further improvement of the foregoing techniques.

(5) In an aspect, the disclosure provides an image forming apparatus including an image forming device, a sheet transport mechanism, a sheet abnormality detection device, an output tray, an escape tray, a notification device, and a control device. The image forming device forms an image on a sheet. The sheet transport mechanism transports the sheet to the image forming device, the sheet transport mechanism including a sheet transport route, and a transport device provided on the sheet transport route and configured to transport the sheet. The sheet abnormality detection device detects abnormality of the sheet, including multi feed and a state of partially missing. The output tray receives the sheet transported by the sheet transport mechanism and on which an image has been formed by the image forming device. The escape tray is provided separately from the output tray, and receives the sheet, from which abnormality has been detected by the sheet abnormality detection device, and delivered by the sheet transport mechanism. The notification device announces sheet abnormality. The control device includes a processor, and controls an overall operation of the image forming apparatus. The control device causes the image forming device to again form, on another sheet, a same image formed on a sheet, from which the sheet abnormality has been detected, and delivered to the escape tray, and to continue with the image forming operation on subsequent sheets, and decides, according to a predetermined condition, whether to allow the notification device to announce the sheet abnormality.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram showing a configuration of an image forming apparatus according to an embodiment of the disclosure;
- (2) FIG. 2 is a flowchart showing an operation performed by the image forming apparatus shown in FIG. 1;
- (3) FIG. 3 is a flowchart showing a process of notifying a content of an escape tray to a user and subsequent operation, to be performed when the user is present in the proximity of the image forming apparatus, or when sheet jam has occurred;
- (4) FIG. 4 is a flowchart showing a process of notifying the content of the escape tray to the user, with priority given to image forming operation, to be performed when the user is present within a predetermined range from the image forming apparatus, or when sheet jam has occurred;
- (5) FIG. 5 is a flowchart showing a process of notifying the content of the escape tray to the user, with the image forming operation suspended, to be performed when the user is present within the predetermined range from the image forming apparatus, or when sheet jam has occurred; and
- (6) FIG. 6 is a flowchart showing an operation after recovery from a sleep mode, performed when the image forming apparatus has a sleep function.

DETAILED DESCRIPTION

- (7) Hereafter, an image forming apparatus according to an embodiment of the disclosure will be described, with reference to the drawings. In the drawings, the same or corresponding elements are given the same numeral, and the description of such elements will not be repeated.
- (8) [Image Forming Apparatus]
- (9) FIG. 1 is a block diagram showing a configuration of the image forming apparatus, including a sheet transport device, according to the embodiment of the disclosure.
- (10) The image forming apparatus 1 shown in FIG. 1 includes an adsorbing transport belt 11 for transporting a sheet 10. The adsorbing transport belt 11 passes an image forming device 13 and a sheet abnormality detection device 14, carrying thereon the sheet 10 delivered from a sheet cassette 12. Here, although FIG. 1 illustrates an example where the image forming device 13 and the sheet abnormality detection device 14 are aligned in this order, along a transport direction of the adsorbing transport belt 11, the image forming device 13 and the sheet abnormality detection

device **14** may be aligned in the reverse order, along the transport direction of the adsorbing transport belt **11**. A sheet transport route is provided, in the section between the sheet cassette **10** and the image forming device **13**, and between the image forming device **13** and an output tray **15** or an escape tray **16**. The sheet transport route includes transport roller pairs that transport the sheet. The adsorbing transport belt **11** may instead be some transport roller pairs. The adsorbing transport belt **11** and the transport roller pairs exemplify the transport device in the disclosure. (11) The image forming device **13** forms an image on the sheet **10**, transported from the sheet cassette **12** by the transport device such as the adsorbing transport belt **11**. The image forming method is not specifically limited, and one of generally known image forming methods may be employed.

(12) The sheet abnormality detection device **14** serves to detect sheet abnormality, for example by detecting multi feed of the sheets **10** adjacent to each other, and/or a partially missing portion of the sheet **10**. The sheet abnormality detection device **14** may be constituted, for example, of a known multi feed sensor, or an image processor that detects a partially missing portion of the sheet, through an image processing operation.

(13) The image forming apparatus **1** shown in FIG. **1** includes the output tray **15**. The output tray **15** serves as the delivery destination for the sheet **10** having thereon an image formed by the image forming device **13**, and from which no sheet abnormality has been detected by the sheet abnormality detection device **14**. The image forming apparatus **1** shown in FIG. **1** also includes the escape tray **16**. The escape tray **16** serves as the delivery destination for the sheet **10** that has undergone the image forming operation by the image forming device **13**, and the detection of sheet abnormality by the sheet abnormality detection device **14**, through which the sheet abnormality has been detected.

(14) The selection of the delivery destination is performed by a branching device **17**. The branching device **17** includes, for example, a known branching nail for shifting the transport route of the sheet **10**. The operation of the branching device **17** is controlled by a controller **18**. The controller **18** exemplifies the control device in the disclosure.

(15) The image forming apparatus **1** shown in FIG. **1** includes the controller **18** for controlling each of the components of the sheet transport device. The controller **18** includes a processor, such as a central processing unit (CPU) or a micro processing unit (MPU).

(16) Between the controller **18** and the image forming device **13**, a control signal line **19** for controlling the operation of the image forming device **13**, with the controller **18**, is provided. Between the sheet abnormality detection device **14** and the controller **18**, an abnormality detection signal line **20** is provided.

(17) A sheet sensor **21** that detects whether a sheet, from which abnormality has been detected, is located on the escape tray **16**, is provided on the escape tray **16**, or in the proximity thereof. The sheet sensor **21** is, for example, an optical sensor having a light emitter and a photodetector. The sheet sensor **21** and the controller **18** are connected to each other via a sheet presence detection signal line **22**, for transmitting a detection signal from the sheet sensor **21** to the controller **18**. The sheet sensor **21** serves to ensure that the presence of the sheet, from which abnormality has been detected, on the escape tray **16**, can be detected without fail.

(18) Between the controller **18** and the branching device **17**, a switching action control signal line **23**, for activating the branching device **17** with the controller **18**, is provided.

(19) The controller **18** receives power from a power source **24**. Although details are not illustrated, the power source **24** serves to supply power to each of the components of the image forming apparatus **1** shown in FIG. **1**, in addition to the controller **18**. The power source **24** includes an on/off switch, for turning on and off the operation of the power source **24**.

(20) To the controller **18**, an operation device **25**, a display device **26**, a human sensor **27**, an open/close sensor **28** for the cover of the apparatus, and a sheet jam sensor **29** are connected via signal transmission lines **30**, **31**, **32**, **33**, and **34**, respectively.

(21) The operation device **25** includes hard keys such as a start key and numeric keys, and receives inputs of instructions corresponding to those keys. The display device **26** includes, for example, a liquid crystal display (LCD). The human sensor **27** is, for example, a known infrared sensor, and detects the user located within a predetermined range (e.g., 1 meter) from the image forming apparatus **1**. The open/close sensor **28** includes a mechanical switch, to be turned on by the pressure of the cover when the cover is closed, and turned off by being released from the pressure of the cover, when the cover is opened. The sheet jam sensor **29** is, for example, an optical sensor having a light emitter and a photodetector, provided at each of predetermined positions on the sheet transport route, to detect whether the sheet **10** is located between the light emitter and the photodetector.

(22) [Operation]

(23) Hereunder, operations of the image forming apparatus **1** and the sheet transport device shown in FIG. **1** will be described. When an image is to be formed on the sheet **10**, the user inputs an instruction to form the image, through the operation device **25**. Then the controller **18** starts to control the action of the sheet transport mechanism, including the transport device such as the adsorbing transport belt **11**. The sheet transport mechanism transports one of the sheets **10** drawn out from the sheet cassette **12**. The image is formed on the surface of the sheet **10** being transported by the sheet transport mechanism, by the image forming device **13**.

(24) The sheet **10** then undergoes the detection of abnormality, by the sheet abnormality detection device **14**. The sheet **10** from which no abnormality has been detected is transported to the output tray **15**, by the adsorbing transport belt **11**. The sheet **10** from which abnormality has been detected is transported to the escape tray **16**, instead of the output tray **15**. The switching of the transport destination between the output tray **15** and the escape tray **16** is performed by the branching device **17**, under the control of the controller **18**.

(25) Hereinafter, a sheet from which abnormality has been detected may be referred to as “invalid sheet”, and a sheet from which no abnormality has been detected may be referred to as “valid sheet”.

(26) Here, in the case where, unlike the example described above, the sheet abnormality detection device **14** and the image forming device **13** are aligned in this order, along the transport direction of the sheet **10** by the transport device such as the adsorbing transport belt **11**, the sheet **10** undergoes the detection of abnormality by the sheet abnormality detection device **14**, before undergoing the image forming operation by the image forming device **13**. In such a case, the image forming operation by the image forming device **13** is successively performed, under the control of the controller **18**, when no abnormality of the sheet **10** has been detected, and thereafter the valid sheet, in other words the sheet **10** having the image formed thereon, is delivered to the output tray **15**. When abnormality is detected from the sheet **10**, the sheet **10** from which the abnormality has been detected, in other words the invalid sheet, is delivered to the escape tray **16**, under the control of the controller **18**, without undergoing the image forming operation by the image forming device **13**, in other words remaining blank.

(27) When the abnormality of the sheet **10** is detected by the sheet abnormality detection device **14**, and the sheet **10** having the abnormality is delivered to the escape tray **16** without undergoing the image forming operation by the image forming device **13**, under the control of the controller **18**, the image that was supposed to be formed on such sheet **10**, in other words the invalid sheet, is formed as scheduled on the next sheet **10** transported from the sheet cassette **12**. The next sheet **10** also undergoes the detection of abnormality by the sheet abnormality detection device **14**.

(28) The image forming apparatus **1** shown in FIG. **1** is configured to continue with the image forming operation on the subsequently delivered sheets **10**, under the control of the controller **18**, without immediately notifying the user of the occurrence of abnormality, even when the invalid sheet is detected. Such operation will be described in detail hereunder, with reference to FIG. **2** and FIG. **1**. FIG. **2** is a flowchart showing the operation of the image forming apparatus **1** shown in

FIG. 1.

(29) Referring to FIG. 2, when the instruction to start the image forming operation is received through the operation device **25** (step S-21), the controller **18** performs the image forming operation on the sheet **10** (step S-22). The image forming operation includes causing the image forming device **13** to form the image on the sheet **10**, and causing the sheet abnormality detection device **14** to detect abnormality from the sheet **10**.

(30) When no abnormality is detected by the sheet abnormality detection device **14** from the sheet **10**, the sheet **10** is transported, with the image formed thereon, to the output tray **15**, under the control of the controller **18**. In this case, the sheet sensor **21** provided on the escape tray **16** does not detect the presence of the invalid sheet on the escape tray **16**, at step S-23 (NO at step S-23). The sheet sensor **21** transmits the data indicating that the invalid sheet has not been detected, to the controller **18**.

(31) Then the operation returns to step S-22, where the image forming operation is performed on the next sheet **10**, under the control of the controller **18**, including the image forming on the next sheet **10** by the image forming device **13**, and the detection of abnormality by the sheet abnormality detection device **14**, from the sheet **10**.

(32) When it is decided that the sheet abnormality detection device **14** has detected abnormality from the sheet **10**, such sheet **10**, in other words the invalid sheet, is transported to the escape tray **16**, under the control of the controller **18**. In this case, the sheet sensor **21** provided on the escape tray **16** detects the presence of the invalid sheet on the escape tray **16** at step S-23 (YES at step S-23), and transmits the detection data to the controller **18**.

(33) The controller **18** decides, at step S-24, whether the presence of the invalid sheet is scheduled to be notified. For example, in the case where the user inputs in advance the instruction to output the notice of the presence of the invalid sheet, through the operation device **25**, the controller **18** stores such instruction in a memory, to make the mentioned decision. Alternatively, the notice is to be outputted on the basis of some other operations, the detail of which will be subsequently described. In the case where the outputting of the notice is not scheduled (NO at step S-24), the operation returns to step S-22, where the image forming operation is performed on the next sheet **10**, under the control of the controller **18**, including the image forming on the next sheet **10** by the image forming device **13**, and the detection of abnormality by the sheet abnormality detection device **14**, from the sheet **10**.

(34) In the case where the presence of the invalid sheet is scheduled to be notified at step S-24 (YES at step S-24), the controller **18** notifies the user, only in such a case, of the presence of the invalid sheet on the escape tray **16**, for example by causing the display device **26** to display a message to the effect that the invalid sheet is present on the escape tray **16** (step S-25). In the image forming apparatus **1** shown in FIG. 1, the display device **26** controlled by the controller **18** serves as a notification device of the presence of the invalid sheet. The notification device may be constituted of an optionally selected device other than the display device **26**, provided that such device is capable to outputting the mentioned notice. After step S-25, the controller **18** suspends the image forming on the sheet **10** by the image forming device **13**, and the detection of abnormality from the sheet **10** by the sheet abnormality detection device **14**. Here, the controller **18** may cause, after step S-25, the image forming device **13** to continue to form the same image that was formed on the sheet **10** delivered to the escape tray **16** because of the sheet abnormality, on another sheet, and cause the sheet abnormality detection device **14** to continue to detect abnormality from the sheet **10**.

(35) In this embodiment, as described above, the controller **18** causes the image forming device **13** to again form, on another sheet, the same image that was formed on the sheet **10** delivered to the escape tray **16** because of the sheet abnormality, and to continue with the image forming operation on the subsequent sheets, and decides, according to a predetermined condition, whether to cause the display device **26** to announce the sheet abnormality.

(36) The mentioned arrangement enables the image forming operation on the sheet **10** to be continued, in the event that the sheet abnormality has been detected, in other words that the invalid sheet has appeared, without notifying the user of such event. Thereafter, the event can be notified to the user, at the time that the notification of the presence of the invalid sheet is requested, in which case the collection work of the invalid sheets by the user can be minimized, and consequently the downtime of the apparatus can be minimized.

(37) [Notice to User and Operation]

(38) Hereunder, some conditions that allow the notification device to notify the user of the presence of the invalid sheet on the escape tray **16** will be described. For example, when the user is located within a predetermined range (e.g., 1 meter) from the image forming apparatus **1**, the controller **18** outputs the notice to the user. In addition, when sheet jam occurs, the controller **18** notifies the user of the presence of the invalid sheet on the escape tray **16**, along with the notification of the occurrence of the sheet jam.

Example 1 of Notice to User and Operation

(39) In this case, it is preferable to output the notice to the user, at a timing other than during the image forming operation on the sheet **10**. As an example of outputting the notice to the user, at a timing other than during the image forming operation on the sheet **10**, FIG. **3** represents a process of notifying the content of the escape tray to the user and subsequent operation, to be performed when the user is located within the predetermined range from the image forming apparatus, or when sheet jam has occurred. Here, the situation where the user is located within the predetermined range from the image forming apparatus **1** encompasses the concept representing the case where the user is located in the proximity of the image forming apparatus **1**.

(40) The flowchart shown in FIG. **3** represents the operation performed when the user is located within the predetermined range from the image forming apparatus **1**, at the timing other than during the image forming operation on the sheet **10**. In this case, the image forming apparatus **1** is on standby ready to perform the image forming on the sheet **10**. Examples of the situation where the user is located within the predetermined range from the image forming apparatus **1** include when the human sensor **27** shown in FIG. **1**, exemplifying the user detection device in the disclosure, has detected the presence of the user, when the power source **24** has been turned on, and when the cover open/close sensor **28**, serving as the user detection device, has detected that the cover of the image forming apparatus **1** has been opened. In addition, the timing other than during the image forming operation on the sheet **10** includes, for example, the timing after the controller **18** has caused the image forming device **13** to execute the image forming operation of a series of images on the sheets **10**, and has caused the sheet transport mechanism to deliver the sheets **10**, on which the series of images have been respectively formed, to the output tray **15** or the escape tray **16**.

(41) For example, when the human sensor **27** shown in FIG. **1** has detected the presence of the user, the controller **18** decides that the user is located within the predetermined range from the image forming apparatus **1** (step S-31). Upon deciding that the user is located within the predetermined range from the image forming apparatus **1**, the controller **18** decides whether the invalid sheet is present on the escape tray **16**, on the basis of the detection signal from the sheet sensor **21** (step S-32).

(42) Upon deciding at step S-32 that the invalid sheet is not present on the escape tray **16**, the controller **18** keeps the image forming apparatus **1** in the standby state, ready to form an image on the sheet **10** (step S-35).

(43) Upon deciding at step S-32 that the invalid sheet is present on the escape tray **16**, the controller **18** transmits a signal indicating the instruction to output the notice, to the display device **26** serving as the notification device, thereby causing the display device **26** to notify the user of the presence of the invalid sheet (step S-33). The method of notifying is the same as in the case of FIG. **2**.

(44) At step S-34, the controller **18** decides whether the user has removed the invalid sheet from the

escape tray **16**, thereby making the escape tray **16** without the invalid sheet. Upon detecting that the invalid sheet is not present, on the basis of the detection signal from the sheet sensor **21** (NO at step S-34), the controller **18** sets the image forming apparatus **1** to the standby state ready to form an image on the sheet **10** (step S-35). In other words, the controller **18** restricts the image forming device **13** from performing the image forming operation, until the sheet from which abnormality was detected is removed from the escape tray and the notice is cancelled, and allows the image forming device **13** to perform the image forming operation, when the sheet from which abnormality was detected is removed from the escape tray and the notice is cancelled. Accordingly, the user is led to remove the invalid sheet from the escape tray **16**, in order to resume the image forming operation, and therefore the sheet from which abnormality was detected can be surely removed from the escape tray **16**.

(45) Upon deciding at step S-34 that the invalid sheet still remains on the escape tray **16**, on the basis of the detection signal from the sheet sensor **21**, the controller **18** returns to step S-33, to continue with notifying the user of the presence of the invalid sheet on the escape tray **16**.

(46) Thus, when the invalid sheet is present on the escape tray **16**, the image forming apparatus **1** shown in FIG. **1** does not resume the operation, unless the user removes the invalid sheet. Therefore, the invalid sheet can be surely removed from the escape tray **16**.

(47) In addition, since the notice to the effect that the invalid sheet is present on the escape tray **16** is outputted when the user is located within the predetermined range from the image forming apparatus **1**, the user can surely be made aware of the presence of the invalid sheet.

(48) Many of existing image forming apparatuses are configured to immediately notify the user of the occurrence of the sheet abnormality, once the sheet abnormality occurs, thereby immediately making the user aware of the occurrence of the abnormality. However, in recent years, in particular in the commercial printing industry, it is becoming more and more important to reduce the downtime per unit time of the apparatus, in order to maintain the productivity. From such a viewpoint, immediately notifying the user of the occurrence of abnormality, once the abnormality occurs, incurs a downtime including suspending of the operation for the purpose of inspection, or collection of the sheet with the abnormality. Moreover, when the user is not located in the proximity of the apparatus, at the time that the notice is outputted, with the operation of the apparatus suspended, the time before the cancellation of the notice is further prolonged.

(49) According to the foregoing embodiment, in contrast, the notice of the occurrence of the sheet abnormality is not immediately outputted once the abnormality occurs, but outputted at an appropriate timing, without interruption of the printing operation, and therefore the downtime of the apparatus can be minimized.

Variation of Operation Example 1

(50) In the foregoing example, the detection of the user located within the predetermined range from the image forming apparatus **1** is performed by the human sensor **27**, serving as the user detection device. Otherwise, the presence of the user within the predetermined range from the image forming apparatus **1** can be detected, for example, when the power source **24** is turned on, and when the cover open/close sensor **28**, serving as the user detection device, detects that the cover of the image forming apparatus **1** has been opened. Such cases are based on a different detection principle as to whether the user is located within the predetermined range from the image forming apparatus **1**, compared with the case of utilizing the detection result from the human sensor **27**. However, the operation performed after the detection is the same as the flowchart shown in FIG. **3**.

Example 2 of Notice to User and Operation

(51) Hereunder, the operation performed to notify the user of the presence of the invalid sheet on the escape tray **16**, along with the notification of the occurrence of the sheet jam, will be described in detail.

(52) Upon detecting the occurrence of sheet jam in the image forming apparatus **1**, on the basis of

the detection signal from the sheet sensor **21**, the controller **18** causes, for example, the display device **26** to notify the user of the occurrence of the sheet jam. The user resolves the sheet jam, in response to the notice. When the invalid sheet is present on the escape tray **16** in this case, notifying the user of the presence of the invalid sheet can urge the user to remove the invalid sheet. In general, the notice of sheet jam is outputted, irrespective of whether the user is present in the proximity of the image forming apparatus **1**. When the sheet jam occurs, the user makes access to the image forming apparatus **1** without fail, to resolve the sheet jam. Accordingly, notifying the user of the presence of the invalid sheet on the escape tray **16** also, irrespective of the detection result as to whether the user is present in the proximity of the image forming apparatus **1**, along with the notice of the occurrence of the sheet jam, enables the user who has made access to the image forming apparatus **1** to be surely notified of those events, thereby motivating the user to both resolve the sheet jam and remove the invalid sheet from the escape tray **16**, at the same timing.

(53) To be more specific, when the sheet jam sensor **29** detects the occurrence of the sheet jam (step S-31 in FIG. 3), the controller **18** starts the operation of step S-32 and the subsequent steps shown in FIG. 3. The operation according to those steps corresponds to “Example 1 of Notice to User and Operation” described earlier.

(54) In the case where the invalid sheet is present on the escape tray **16**, at the time that the sheet jam has occurred, the controller **18** sets the image forming apparatus **1** to the standby state ready to form an image on the sheet **10**, when the user removes the invalid sheet and the sheet sensor **21** detects that the invalid sheet is not present. In other words, the image forming apparatus **1** does not resume the operation despite the sheet jam having been resolved, unless the user removes the invalid sheet. The mentioned arrangement motivates the user to remove the invalid sheet from the escape tray **16**, at the time that the sheet jam has occurred.

Example 3 of Notice to User and Operation

(55) Another example of “notice to user and operation” is allowing the user to select one of (a) giving priority to the image forming on the sheet **10**, to outputting the notice, and (b) giving priority to outputting the notice to the user, despite the image forming on the sheet **10** being performed, for example suspending the series of image forming operations on the sheets **10**, thereby urging the user to remove the invalid sheet from the escape tray **16**.

(56) The case (a) will be described in detail hereunder. In the image forming apparatus **1**, the image forming device **13** is basically configured to continue with the image forming operation on the subsequent sheets **10**, even when sheet abnormality is detected. Further detail will be described with reference to FIG. 4.

(57) FIG. 4 is a flowchart showing the operation performed when the user selects notifying the content of the escape tray to the user, with priority given to the image forming operation, when the user is present within the predetermined range from the image forming apparatus, or when sheet jam has occurred. Referring to FIG. 4, for example when it is detected at step S-41, as in the case of FIG. 3, that the user is located within the predetermined range from the image forming apparatus **1**, or that sheet jam has occurred, during the image forming operation, the controller **18** starts the operation of step S-42 and the subsequent steps. At step S-42, the controller **18** decides whether the invalid sheet is present on the escape tray **16**, on the basis of the detection signal from the sheet sensor **21**. Then the controller **18** completes the series of image forming operations on the sheets **10** at step S-43, irrespective of whether the invalid sheet is present on the escape tray **16**.

(58) Then the controller **18** decides at step S-44 whether the user is present within the predetermined range from the image forming apparatus, or whether sheet jam has occurred. In the case where the user is not present within the predetermined range from the image forming apparatus, and sheet jam has not occurred (NO at step S-44), the controller **18** keeps the image forming apparatus **1** in the standby state ready to form an image on the sheet **10** (step S-48).

(59) Upon deciding at step S-44 that the user is present within the predetermined range from the image forming apparatus, or that sheet jam has occurred, the controller **18** performs the operation

of step S-45 to step S-48. The operation of step S-45 to step S-48 is the same as the operation of step S-32 to step S-35 shown in FIG. 3. Therefore, the description of those steps will not be repeated.

Example 4 of Notice to User and Operation

(60) Referring to FIG. 5, the case of (b), exceptionally giving priority to outputting the notice to the user, despite the image forming on the sheet **10** being performed, for example suspending the series of image forming operations on the sheets **10**, thereby urging the user to remove the invalid sheet from the escape tray **16**, will be described in detail hereunder. FIG. 5 is a flowchart showing the operation performed when the user selects notifying the content of the escape tray to the user, with the image forming operation suspended, when the user is present within the predetermined range from the image forming apparatus, or when sheet jam has occurred.

(61) Referring to FIG. 5, for example when it is detected at step S-51, as in the case of FIG. 3, that the user is located within the predetermined range from the image forming apparatus **1**, or that sheet jam has occurred, during the image forming operation, the controller **18** starts the operation of step S-52 and the subsequent steps. At step S-52, the controller **18** decides whether the invalid sheet is present on the escape tray **16**. Upon deciding at step S-52 that the invalid sheet is not present on the escape tray **16**, the controller **18** keeps the image forming apparatus **1** in the standby state ready to form an image on the sheet **10** (step S-56). The mentioned steps are the same as those of FIG. 3.

(62) Upon deciding at step S-52 that the invalid sheet is present on the escape tray **16**, the controller **18** causes, for example at this point, the image forming device **13** to suspend the image forming operation on the sheet **10** (step S-53). Then the controller **18** notifies the user of the presence of the invalid sheet on the escape tray **16** (step S-543). The notification method is the same as the case of FIG. 2.

(63) After step S-54, the controller **18** performs the operation of step S-55 and step S-56. The operation of step S-54 to step S-56 is the same as the operation of step S-33 to step S-35 shown in FIG. 3. Therefore, the description of those steps will not be repeated.

(64) The controller **18** performs the operation including one of (a) giving priority to the image forming on the sheet **10**, to outputting the notice, and (b) giving priority to outputting the notice to the user, despite the image forming on the sheet **10** being performed, for example suspending the series of image forming operations on the sheets **10**, thereby urging the user to remove the invalid sheet from the escape tray **16**.

(65) Thus, the operations of (a) and (b) can be selectively performed when the invalid sheet appears, and the presence of the invalid sheet on the escape tray **16** can be notified to the user, or kept from being notified to the user, at the time that the invalid sheet has been detected, according to the operation performed by the user on the operation device **25**, in this case serving as the selection device. Allowing thus the user to select one of the operations (a) and (b) enables the image forming apparatus **1** to flexibly operate, under various conditions.

Example 5 of Notice to User and Operation

(66) The image forming apparatus **1** shown in FIG. 1 may be given a sleep function. The sleep function refers to a function to set the apparatus to an energy-saving mode. The controller **18** sets the image forming apparatus **1** to a sleep state, by performing the sleep function.

(67) Referring to FIG. 6, an operation to cancel the sleep mode and recover from the sleep state, performed when the image forming apparatus **1** has the sleep function, will be described hereunder. FIG. 6 is a flowchart showing the operation after recovery from the sleep mode, performed when the image forming apparatus has the sleep function.

(68) Upon waking the image forming apparatus **1** from the sleep mode at step S-61, the controller **18** detects whether the invalid sheet is present on the escape tray **16** (step S-62). The controller **18** does not notify the user of the occurrence of sheet abnormality, irrespective of whether the invalid sheet is present on the escape tray **16**, and decides whether the user is located in the proximity of

the image forming apparatus **1**, or whether sheet jam has occurred (step S-63).

(69) In the case where the user is not located within the predetermined range from the image forming apparatus **1**, and the sheet jam has not occurred, the controller **18** restricts the image forming apparatus **1** from outputting the notice, and from suspending the image forming operation. In contrast, upon deciding at step S-63 that the user is located within the predetermined range from the image forming apparatus **1**, or that the sheet jam has occurred, the controller **18** performs the operation of step S-64 to step S-66. The operation of step S-64 to step S-66 is the same as the operation of step S-33 to step S-35 shown in FIG. 3. Therefore, the description of those steps will not be repeated.

(70) In other words, in the case where the controller **18** once sets the image forming apparatus **1** to the sleep mode, and then cancels the same, the controller **18** restricts the display device **26**, serving as the notification device, from outputting the notice of sheet abnormality, despite the sheet sensor **21** having detected that the sheet **10** is present on the escape tray **16**, and allows the display device **26** to output the notice of the sheet abnormality, when the human sensor **27** detects the presence of the user or when the sheet jam sensor **29** detects the occurrence of sheet jam, after the sleep mode is cancelled. Thereafter, the controller **18** restricts the image forming device **13** from performing the image forming operation, until causing the display device **26** to erase the notice of the sheet abnormality, because of the sheet sensor **21** no longer detecting the presence of the sheet **10** on the escape tray **16**. This is because, while the image forming apparatus **1** is in the sleep mode, the image forming apparatus **1** is kept from being utilized by the user, for example from receiving an instruction inputted by the user through the operation device **25**, and therefore it may not always be beneficial to give the aforementioned motivation to the user during the sleep state, by notifying the sheet abnormality.

(71) With the arrangement according to the foregoing embodiment, therefore, also when the image forming apparatus **1** has the sleep function, the user can be motivated to resolve the sheet jam, and to remove the invalid sheet from the escape tray **16**, at the timing that the sleep mode is cancelled.

(72) The embodiment of the disclosure has been described as above, with reference to the drawings. However, the disclosure is not limited to the foregoing embodiment, but may be modified in various manners, without departing from the scope of the disclosure. The drawings each schematically illustrate the elements for the sake of clarity, and the thickness, length, number of pieces, and interval of the illustrated elements may be different from the actual ones, because of the convenience in making up the drawings. Further, the material, shape, and size of the elements referred to in the foregoing embodiment are merely exemplary and not specifically limited, and may be modified as desired, without substantially departing from the configuration according to the disclosure.

INDUSTRIAL APPLICABILITY

(73) The disclosure provides the image forming apparatus, in particular the sheet transport mechanism installed therein, and is therefore industrially applicable.

(74) While the present disclosure has been described in detail with reference to the embodiments thereof, it would be apparent to those skilled in the art the various changes and modifications may be made therein within the scope defined by the appended claims.

Claims

1. An image forming apparatus comprising: an image forming device that forms an image on a sheet; a sheet transport mechanism that transports the sheet to the image forming device, the sheet transport mechanism including a sheet transport route, and a transport device provided on the sheet transport route and configured to transport the sheet; a sheet abnormality detection device that detects abnormality of the sheet, including multi feed and a state of partially missing; an output tray that receives the sheet transported by the sheet transport mechanism, and on which an image has

been formed by the image forming device; an escape tray that receives the sheet, from which abnormality has been detected by the sheet abnormality detection device, and delivered by the sheet transport mechanism, the escape tray being provided separately from the output tray; a notification device that announces sheet abnormality; and a control device including a processor, and configured to control an overall operation of the image forming apparatus, wherein the control device being configured to: cause the image forming device to again form, on another sheet, a same image formed on the sheet, from which the sheet abnormality has been detected, and delivered to the escape tray, and to continue with image forming operation on subsequent sheets; and decide, according to a predetermined condition, whether to allow the notification device to announce the sheet abnormality, wherein the image forming apparatus further comprises: a sheet sensor that detects whether a sheet is present in the escape tray; and a user detection device that detects a user located within a predetermined range from the image forming apparatus, wherein the control device causes the notification device to announce the sheet abnormality, when the user detection device has detected presence of the user, after causing the image forming device to execute the image forming operation of a series of images on the sheets, and causing the sheet transport mechanism to deliver the sheets subjected to the image forming operation of the series of images to one of the output tray and the escape tray, and when the sheet sensor has detected that the sheet is present on the escape tray.

2. The image forming apparatus according to claim 1, wherein the control device restricts the image forming device, after causing the notification device to announce the sheet abnormality, when the sheet jam sensor detected the occurrence of the sheet jam, and the sheet sensor detected that the sheet is present on the escape tray, from forming an image on the sheet, until the sheet sensor detects that the sheet is no longer present on the escape tray, and allows the image forming device to form the image, when the sheet sensor has detected that the sheet is no longer present on the escape tray.

3. An image forming apparatus comprising: an image forming device that forms an image on a sheet; a sheet transport mechanism that transports the sheet to the image forming device, the sheet transport mechanism including a sheet transport route, and a transport device provided on the sheet transport route and configured to transport the sheet; a sheet abnormality detection device that detects abnormality of the sheet, including multi feed and a state of partially missing; an output tray that receives the sheet transported by the sheet transport mechanism, and on which an image has been formed by the image forming device; an escape tray that receives the sheet, from which abnormality has been detected by the sheet abnormality detection device, and delivered by the sheet transport mechanism, the escape tray being provided separately from the output tray; a notification device that announces sheet abnormality; and a control device including a processor, and configured to control an overall operation of the image forming apparatus, wherein the control device being configured to: cause the image forming device to again form, on another sheet, a same image formed on the sheet, from which the sheet abnormality has been detected, and delivered to the escape tray, and to continue with image forming operation on subsequent sheets; and decide, according to a predetermined condition, whether to allow the notification device to announce the sheet abnormality, wherein the image forming apparatus further comprises: a sheet sensor that detects whether a sheet is present in the escape tray; and a sheet jam sensor that detects occurrence of sheet jam on the sheet transport route, wherein the control device causes the notification device to announce the sheet abnormality, when the sheet jam sensor has detected the occurrence of the sheet jam, and when the sheet sensor has detected that the sheet is present on the escape tray.

4. An image forming apparatus comprising: an image forming device that forms an image on a sheet; a sheet transport mechanism that transports the sheet to the image forming device, the sheet transport mechanism including a sheet transport route, and a transport device provided on the sheet transport route and configured to transport the sheet; a sheet abnormality detection device that detects abnormality of the sheet, including multi feed and a state of partially missing; an output tray

that receives the sheet transported by the sheet transport mechanism, and on which an image has been formed by the image forming device; an escape tray that receives the sheet, from which abnormality has been detected by the sheet abnormality detection device, and delivered by the sheet transport mechanism, the escape tray being provided separately from the output tray; a notification device that announces sheet abnormality; and a control device including a processor, and configured to control an overall operation of the image forming apparatus, wherein the control device being configured to: cause the image forming device to again form, on another sheet, a same image formed on the sheet, from which the sheet abnormality has been detected, and delivered to the escape tray, and to continue with image forming operation on subsequent sheets; and decide, according to a predetermined condition, whether to allow the notification device to announce the sheet abnormality, wherein the image forming apparatus further comprises: a sheet sensor that detects whether a sheet is present in the escape tray; and an operation device through which an instruction from the user is inputted, wherein, when presence of the sheet on the escape tray is detected by the sheet sensor, while the image forming device is forming an image on the sheet, the control device is configured to selectively perform, according to the instruction inputted through the operation device, one of: a control including temporarily suspending the image forming operation of a series of images on the sheets by the image forming device, and causing the notification device to announce the presence of the sheet on the escape tray, when the user detection device has detected presence of the user, or when the sheet jam sensor has detected occurrence of sheet jam; and a control including restricting the notification device from announcing the presence of the sheet on the escape tray, while the image forming device is forming an image on the sheet, and allowing the notification device to announce the presence of the sheet on the escape tray, after the image forming device has completed the image forming operation of the series of images on the sheets.

5. An image forming apparatus comprising: an image forming device that forms an image on a sheet; a sheet transport mechanism that transports the sheet to the image forming device, the sheet transport mechanism including a sheet transport route, and a transport device provided on the sheet transport route and configured to transport the sheet; a sheet abnormality detection device that detects abnormality of the sheet, including multi feed and a state of partially missing; an output tray that receives the sheet transported by the sheet transport mechanism, and on which an image has been formed by the image forming device; an escape tray that receives the sheet, from which abnormality has been detected by the sheet abnormality detection device, and delivered by the sheet transport mechanism, the escape tray being provided separately from the output tray; a notification device that announces sheet abnormality; and a control device including a processor, and configured to control an overall operation of the image forming apparatus, wherein the control device being configured to: cause the image forming device to again form, on another sheet, a same image formed on the sheet, from which the sheet abnormality has been detected, and delivered to the escape tray, and to continue with image forming operation on subsequent sheets; and decide, according to a predetermined condition, whether to allow the notification device to announce the sheet abnormality, wherein the image forming apparatus further comprises a sheet sensor that detects whether a sheet is present in the escape tray, wherein the control device is configured to: set the image forming apparatus to a sleep mode; restrict the notification device, when the sleep mode is cancelled, from announcing the sheet abnormality, despite the sheet sensor having detected the presence of the sheet on the escape tray; allow the notification device to announce the sheet abnormality, when the user detection device has detected the presence of the user, or the sheet jam sensor has detected occurrence of sheet jam, after the sleep mode is cancelled; and restrict the image forming device, after the sleep mode is cancelled and after the notification device announced the sheet abnormality, from forming an image, until causing the notification device to stop announcing the sheet abnormality, because of the sheet sensor no longer detecting the presence of the sheet on the escape tray.

